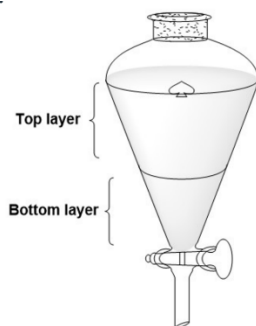


Chem 0345/0330 – Organic Lab

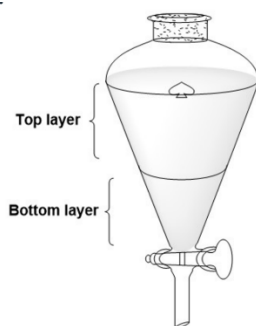
Activity 3: Extraction

Extra Practice Problems

1. Suppose there is a 100 mL solution of bromoaniline ($\text{C}_6\text{H}_5\text{BrN}$) in butanone ($\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$, $\rho = 0.81 \text{ g/mL}$). To the solution is added 100 mL of 1 M $\text{HCl}_{(\text{aq})}$ and the combined mixture is added to a separatory funnel.



- A. Which layer is the butanone?
B. Which layer is the water?
C. In what form is bromoaniline?
D. In which layer is the bromoaniline is mostly distributed?
2. Suppose there is a 100 mL solution of caproic acid ($\text{CH}_3(\text{CH}_2)_4\text{CO}_2\text{H}$) in butanone ($\text{CH}_3\text{C}(\text{O})\text{CH}_2\text{CH}_3$, $\rho = 0.81 \text{ g/mL}$). To the solution is added 100 mL of 1 M $\text{KOH}_{(\text{aq})}$ and the combined mixture is added to a separatory funnel.



- A. Which layer is the butanone?
B. Which layer is the water?
C. In what form is caproic acid?
D. In which layer is the caproic acid is mostly distributed?

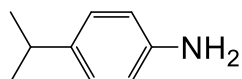
3. Water ($\rho = 1.0 \text{ g/mL}$) and methanol ($\rho = 0.79 \text{ g/mL}$) are mixed in a separatory funnel. Which solvent will be the bottom layer?
 - A. Water
 - B. Methanol
 - C. Neither; they are miscible and will form a single layer

4. Water ($\rho = 1.0 \text{ g/mL}$) and toluene ($\rho = 0.87 \text{ g/mL}$) are mixed in a separatory funnel. Which solvent will be the top layer?
 - A. Water
 - B. Toluene
 - C. Neither; they are miscible and will form a single layer

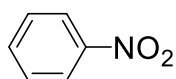
5. Water ($\rho = 1.0 \text{ g/mL}$) and chlorobenzene ($\rho = 1.11 \text{ g/mL}$) are mixed in a separatory funnel. Which solvent will be the bottom layer?
 - A. Water
 - B. Chlorobenzene
 - C. Neither; they are miscible and will form a single layer

6. Water ($\rho = 1.0 \text{ g/mL}$) and isopropanol ($\rho = 0.66 \text{ g/mL}$) are mixed in a separatory funnel. Which solvent will be the top layer?
 - A. Water
 - B. Isopropanol
 - C. Neither; they are miscible and will form a single layer

7. Which of the following statements is true about the two molecules below?



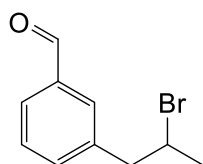
A



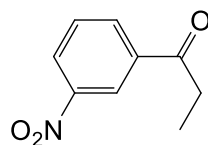
B

- A. Molecule A and B can only be separated by a base extraction
 - B. Molecule A and B can only be separated by an acid extraction
 - C. Molecule A and B can be separated by both a base and an acid extraction
 - D. Molecule A and B cannot be separated by either a base extraction or an acid extraction

8. Which of the following statements is true about the two molecules below?



A



B

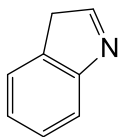
- A. Molecule A and B can only be separated by a base extraction
 - B. Molecule A and B can only be separated by an acid extraction

- C. Molecule A and B can be separated by both a base and an acid extraction
D. Molecule A and B cannot be separated by either a base extraction or an acid extraction

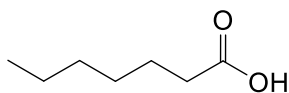
9. Calculate the distribution coefficient (K_D) of compound X if the concentration of X is 10 M in chlorobenzene ($\rho = 1.11 \text{ g/mL}$) and 4 M in the aqueous layer. Which layer will contain most of the compound?

10. Calculate the distribution coefficient (K_D) of compound X if the concentration of X is 2 M in toluene ($\rho = 0.87 \text{ g/mL}$) and 4 M in the aqueous layer. Which layer will contain most of the compound?

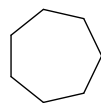
11. Rank the following molecules from least to most polar.



A

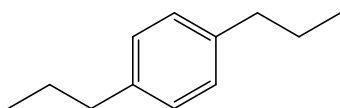


B

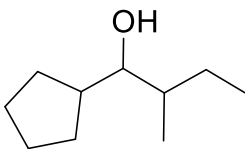


C

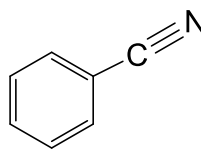
12. Rank the following molecules from most to least polar.



A

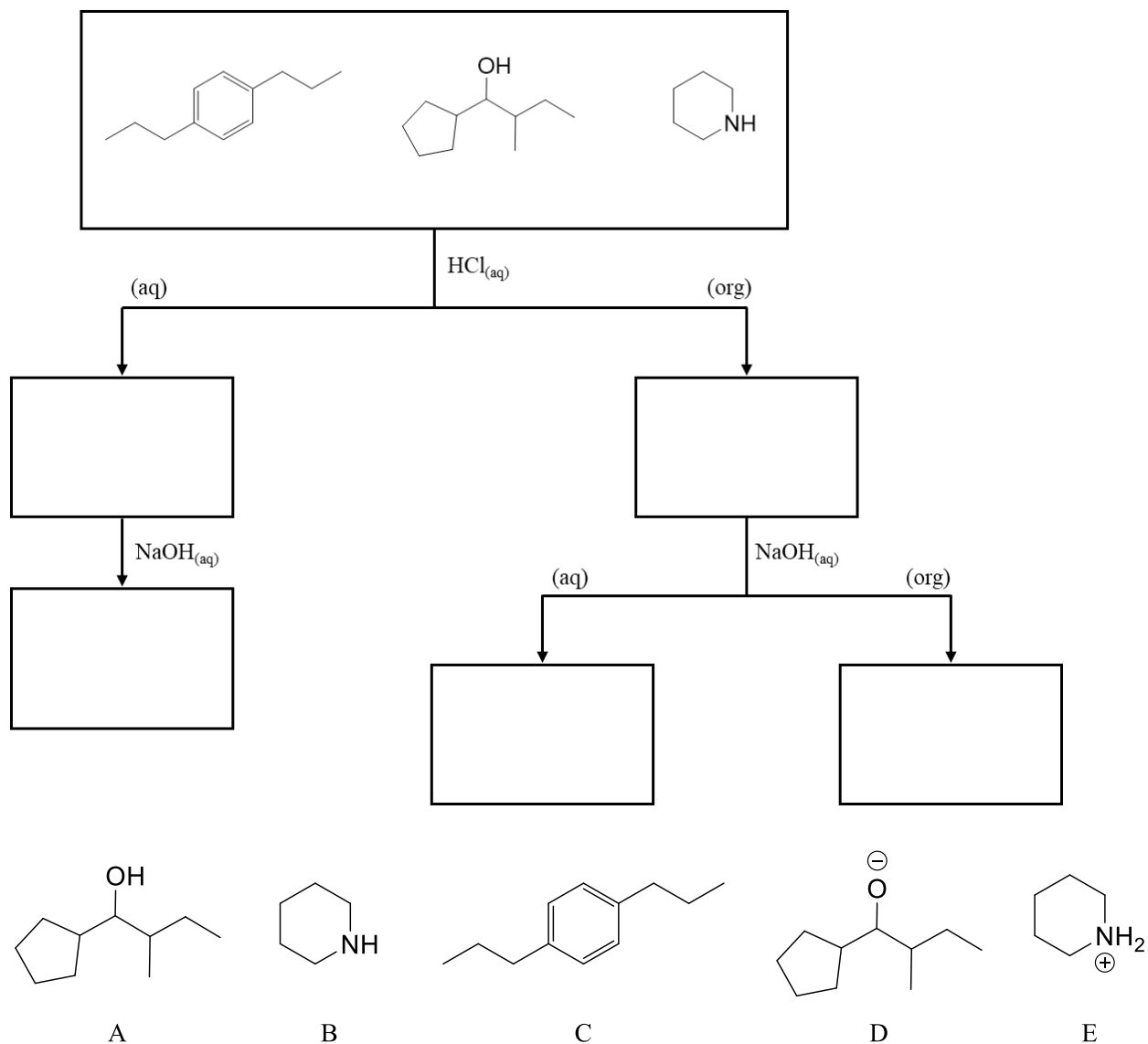


B



C

13. Given a 1 L solution of diethyl ether containing ~1 g of each of the following solutes, fill in the blanks for the following extraction scheme to **SEPARATE** each compound (there is no need to provide purification of each compound, only separation from the others). The aqueous or organic for each phase is labelled. Assume there is no reaction between the compounds themselves.



The mixture contains: c1ccc2c(c1)c[nH]2 (Indole), CCCCCCCC(=O)O (Heptanoic acid), and C1CCCCC1=O (Cycloheptanone).

Reagents and steps:

- Add $\text{NaOH}_{(\text{aq})}$.
 - Organic layer (org): c1ccc2c(c1)c[nH]2 (Indole)
 - Aqueous layer (aq): CCCCCCCC(=O)[O-] (Heptanoate)
- Treat the organic layer with $\text{HCl}_{(\text{aq})}$.
 - Organic layer (org): c1ccc2c(c1)c[nH]2 (Indole, D)
 - Aqueous layer (aq): CCCCCCCC(=O)O (Heptanoic acid, C)
- Treat the aqueous layer from step 1 with $\text{HCl}_{(\text{aq})}$.
 - Aqueous layer (aq): CCCCCCCC(=O)O (Heptanoic acid, C)

Final products:

- A: CCCCCCCC(=O)[O-] (Heptanoate)
- B: c1ccc2c(c1)c[nH+]2 (Indole cation)
- C: CCCCCCCC(=O)O (Heptanoic acid)
- D: c1ccc2c(c1)c[nH]2 (Indole)
- E: C1CCCCC1=O (Cycloheptanone)

